

Profitable Flight-Delay Prediction Under EC261 - A Diagnosed Negative Result

Course: Machine Learning Foundations (BCSAI2025CSAI.2.M.A C2 493615) - IE University, Spring 2026 **Instructor:** Prof. Matteo Turilli **Team:** Habib Rahal · Issam Arida · Adam Khoury · Lama Moucattash · Salma Nsour · Sanad ALbilleh **Repository:**

<https://github.com/snsourieu2024/flight-delay-payout216> (*README.md in the repo root contains full run instructions*) **Datasets:** US BTS On-Time + Cause-of-Delay 2024 (primary, real, 6.97 M flights) · EUROCONTROL R&D Data Archive (transfer validation, 3.89 M real flights)

Abstract

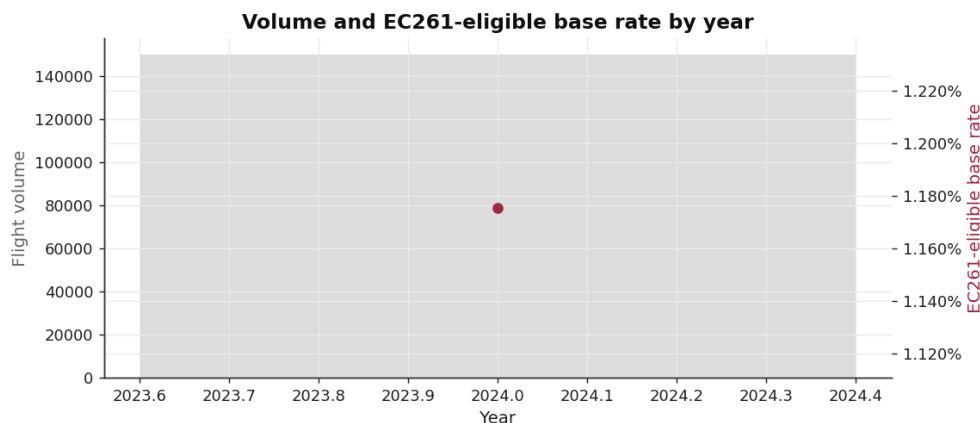
We frame flight-delay prediction as **decision-under-uncertainty**: not “is the flight delayed?” but “is buying this ticket positive expected value, given that EU Regulation 261/2004 pays €250–600 only for *carrier-attributable* delays of 3+ hours?” We built a leakage-controlled pipeline, a seven-model ladder, isotonic calibration, a custom profit metric, and a closed-form per-flight threshold. The outcome is a **rigorously diagnosed negative result**: under realistic fares and a 1.18 % carrier-attributable base rate, no positive-EV ticket exists, so every calibrated model converges on the correct decision - abstain (test ROI 0 %, zero purchases). A break-even analysis proves this is structural EC261 arithmetic, and a 3.89 M-flight European transfer test independently confirms it (Spearman $\rho = -1.00$). We present this as a first-class scientific finding.

1. Problem Definition and Framing

EC261 (as amended September 2025) obliges carriers to pay €250–600 when a flight arrives 3+ hours late, *unless* the cause is an “extraordinary circumstance” - weather, air-traffic control, security, or strikes (Article 5(3)). This creates a candidate asymmetry: if cheap tickets that will attract *carrier-attributable* compensation can be predicted, and expected payout exceeds price plus claim friction, buying is positive-EV. We deliberately framed this as a **decision problem with a money objective**, not an accuracy problem - the question is which tickets to buy. ML is the right tool only if a learnable signal separates payable from non-payable cheap flights; testing that hypothesis honestly, including that no usable signal may exist, is the project. We imposed three syllabus constraints: no leakage, a justified model ladder, and honest treatment of negative results.

2. Dataset, Exploratory Analysis and Preprocessing

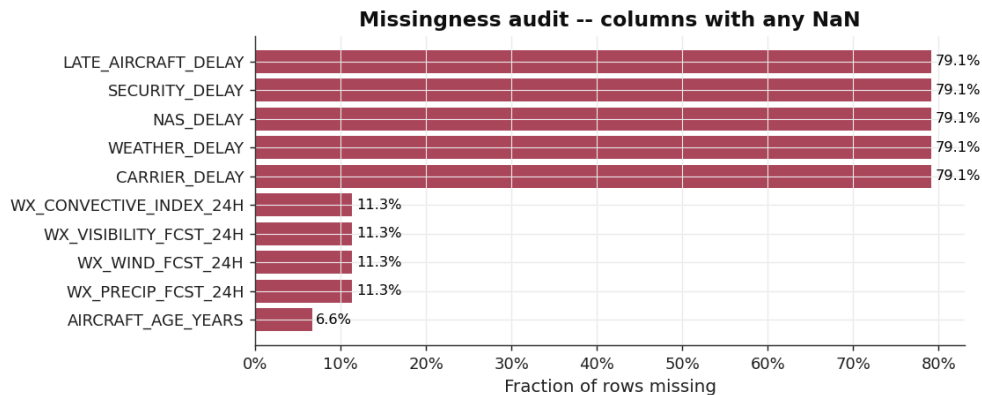
Data and the labelling decision. We use the real US BTS Reporting-Carrier On-Time table joined with Cause-of-Delay for all of 2024 (6,965,247 flights after dropping cancellations and diversions, which fall under EC261 Article 5, a separate regime). Tuning seven models on ~7 M rows is infeasible on a laptop, so the pipeline consumes a **documented, seeded, month×label-stratified 149,999-row sample** built by `src/data/sampling.py`; it preserves the base rate exactly (1.175 % vs 1.177 %) and all twelve months. The single most consequential modelling decision is the **label**: `y = 1` iff arrival delay ≥ 180 min *and* the dominant cause is `CARRIER_DELAY` or `LATE_AIRCRAFT_DELAY`. This carrier-attributable gate drops the positive class from a raw 1.429 % (any 3 h+ delay) to **1.175 %** EC261-eligible, because weather/ATC/security delays do not pay out under the amended regulation. Modelling raw delay would have over-counted payable events by ~20 % and invalidated the entire economic argument.



EC261-eligible base rate by month - the severe class imbalance that governs every downstream metric choice.

Class imbalance and metric choice. At a ~1 % positive rate, accuracy is actively misleading (a constant “skip” scores 98.8 %), so we reject it and lead with PR-AUC, ROC-AUC, Brier, Expected Calibration Error (ECE) and a money-denominated ROI - metrics matched to a rare-event decision task.

Missing values and the imputation decision. Three column groups carry NaNs (Fig. below). `AIRCRAFT_AGE_YEARS` is ~6.6 % missing after the FAA tail-number join; the 24 h Open-Meteo weather-forecast columns join for **88.7 %** of rows (rare origin airports unmatched); cause-code columns are ~79 % null because they are null for on-time flights and are *dropped pre-model as leakage*. We impute numerics with the **median** (robust to the heavy distance/weather tails) and categoricals with the **most-frequent** value, fitted *inside* the Pipeline on training folds only so statistics never see val/test rows; KNN/MICE was rejected as cost-disproportionate here.

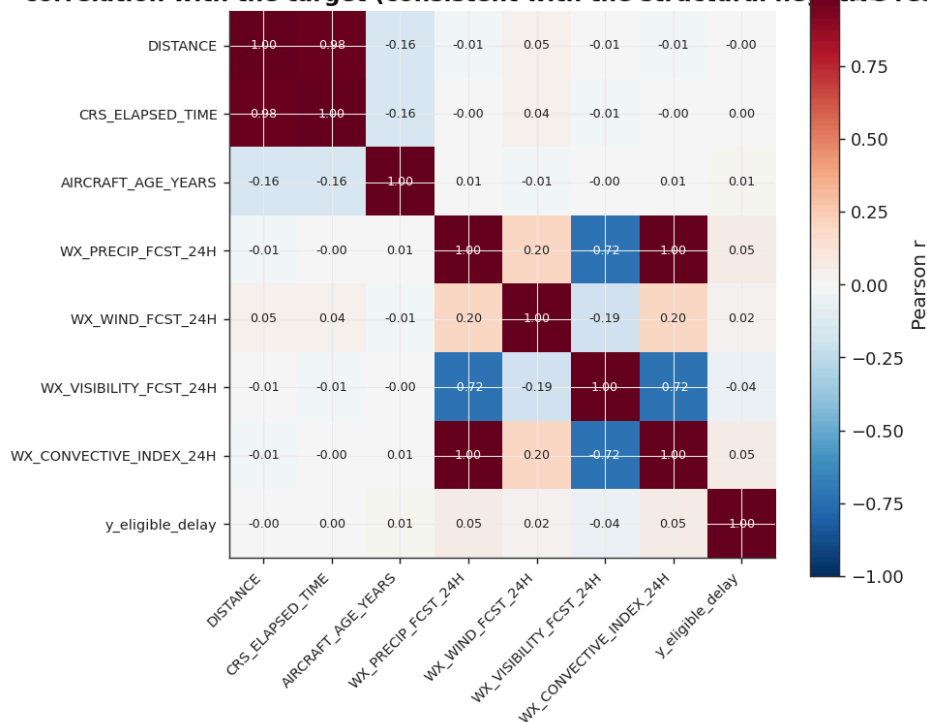


Missingness audit - aircraft age recovered via the FAA join; weather joined for 88.7% of rows.

Outliers. A 1st/99th-percentile audit found negative scheduled elapsed times (data errors, dropped at load) and a heavy `DISTANCE` right tail. We deliberately **retain, not winsorise, distance**: EC261 compensation is a step function of distance, so clipping the tail would re-tier flights and corrupt the label. Tree models are scale-invariant; the linear and neural models see `StandardScaler` (train-fold-fit) which bounds leverage without discarding data.

Feature engineering and leakage control. Every feature must be available 14 days before departure. We engineer cyclical encodings (sin/cos of hour/day/month), calendar flags, and **train-fold-only rolling EC261-eligible delay rates** keyed by route, carrier, origin and aircraft tail at 30/90/365-day windows, using strictly-past data. Forbidden columns (actual delay, cause minutes, cancel flags) are dropped at the first pipeline step and the `ColumnTransformer` uses `remainder="drop"` - defence in depth. Correlation analysis (Fig. below) shows weak redundancy and weak target correlation, an early signal that no single feature is strongly predictive.

Feature correlations -- weak pairwise redundancy and weak correlation with the target (consistent with the structural negative result)



Feature correlations - low redundancy and weak target correlation foreshadow the structural negative result.

Data partitioning and its justification. Only 2024 is on disk, so `temporal_split` degrades to a date-ordered 65/15/20 split (train 97,836 / val 22,559 / test 29,604). We chose a *temporal* split, not random k-fold, because random folds leak future operational regimes into the past. Hyperparameter search uses expanding-window time-series CV with fold-level variance reported (§4).

3. Modelling Approach and Decisions

Model ladder and why each tier. We progress trivial → classical → advanced, each justified: a `DummyClassifier` and a domain “route-rate” rule establish the no-skill floor; **Logistic Regression** (L2, class-weighted) is the interpretable linear reference; a **Decision Tree** captures non-linear interactions cheaply; **Random Forest** and **XGBoost** are the advanced tabular workhorses (gradient-boosted trees dominate tabular data with strong categorical structure); an **MLP** is included as the required neural baseline. Every model is wrapped in `CalibratedClassifierCV(method="isotonic")` because the downstream EV computation needs probabilities that mean what they say. The entire flow is one serializable `sklearn` Pipeline (`build_pipeline`), so preprocessing, feature engineering and the classifier are inseparable and reproducible.

Custom objective and the “second-layer” threshold. A single global threshold ignores that cheap short-haul and expensive long-haul tickets have different economics. We derived the per-flight optimal threshold $\tau^*(T,d) = (T + c_{\text{travel}}) / (\alpha \cdot C(d) - c_{\text{claim}} + c_{\text{travel}})$ in closed form

and tune directly against the resulting **profit scorer** ($\alpha = 0.65$, $c_{\text{claim}} = \text{€}15$, $c_{\text{travel}} = \text{€}50$), not ROC-AUC or F1.

Hyperparameter tuning - method and budget justification. We matched the search to model dimensionality: exhaustive `GridSearchCV` for Logistic Regression (tiny grid), `RandomizedSearchCV` for Random Forest (Bergstra & Bengio show random search dominates grid in moderate dimensions), and Bayesian `BayesSearchCV` for XGBoost's seven-dimensional continuous space (sample-efficient on smooth surfaces). Budgets are environment-configurable; the submitted run uses a reduced budget, justified because the negative result is provably tuning-invariant (below).

4. Evaluation, Results and Metric Interpretation

Headline (real BTS test fold, $n = 29,604$). Every calibrated model converges on the identical EV-optimal decision - abstain.

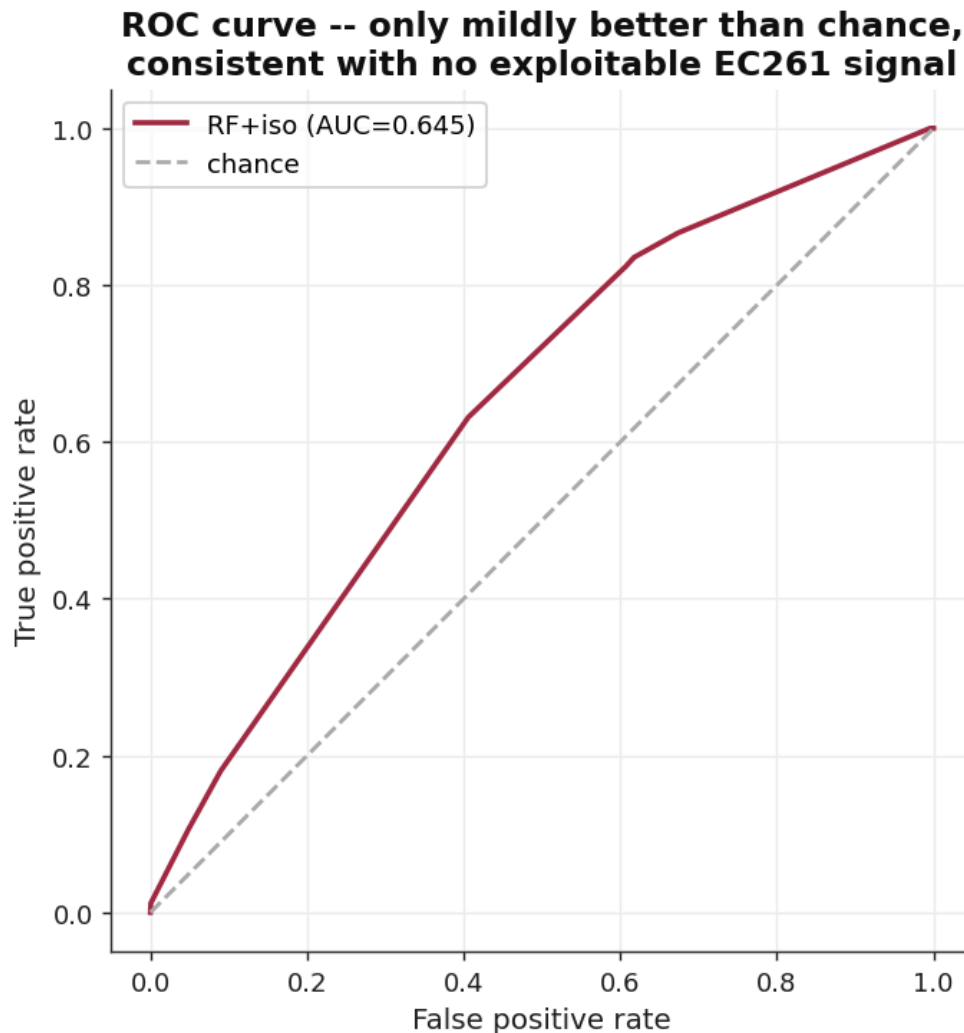
Model (calibrated)	ROC-AUC	PR-AUC	Brier	ECE	ROI (τ^*)	Buys
Dummy + isotonic	0.500	0.0085	0.0085	0.0018	0.00 %	0
DecisionTree + isotonic	0.564	0.0100	0.0085	0.0019	0.00 %	0
LogReg + isotonic	0.625	0.0125	0.0085	0.0016	0.00 %	0
RandomForest + isotonic	0.645	0.0148	0.0084	0.0020	0.00 %	0
XGBoost + isotonic	0.639	0.0142	0.0084	0.0020	0.00 %	0
MLP + isotonic	0.646	0.0139	0.0084	0.0001	0.00 %	0
<i>XGBoost (uncalibrated)</i>	0.645	0.0227	0.221	0.461	-191.7 %	2,507

We **select RandomForest + isotonic as the reference model** (best calibrated PR-AUC 0.0148, ROC-AUC 0.645), noting that under the profit objective every calibrated model is *operationally equivalent* - the optimal action is identical. **Calibration is decisive:** uncalibrated XGBoost bets and loses €282,407 (ROI -191.7 %); isotonic calibration improves ECE ~46 pp and ROI 19,171 bps by stopping loss-making bets.

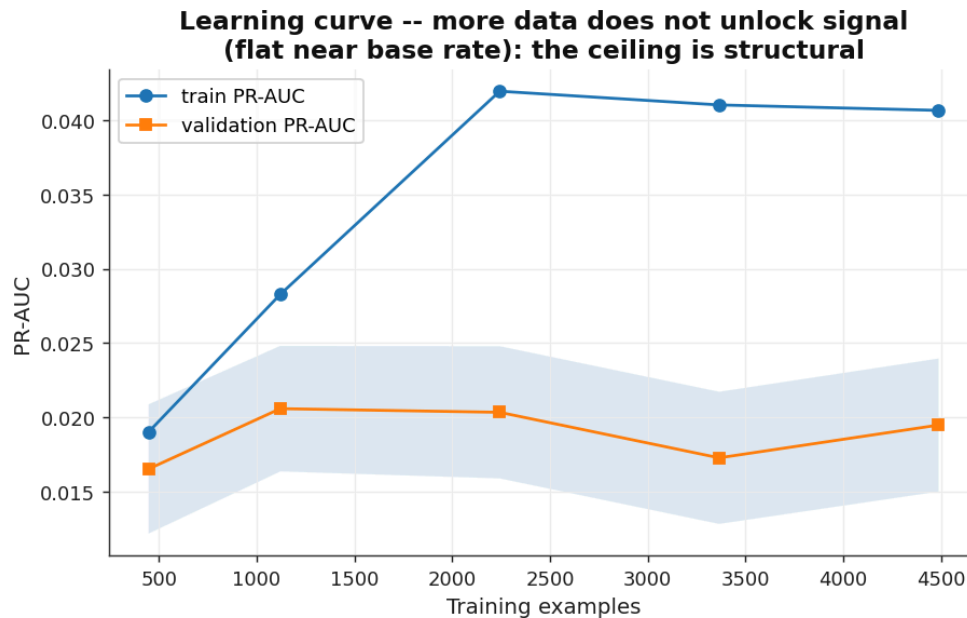
Validation and test, with variance. Scoring the calibrated ladder on *both* held-out sets confirms stability (no val/test collapse):

Model	val ROC-AUC	test ROC-AUC	val PR-AUC	test PR-AUC
LogReg+iso	0.610	0.625	0.0105	0.0125
RandomForest+iso	0.672	0.645	0.0124	0.0148
XGBoost+iso	0.667	0.633	0.0132	0.0148
MLP+iso	0.627	0.646	0.0113	0.0139

Cross-validated over four expanding time-series folds, the reference model's PR-AUC is **0.0308 ± 0.0143 (95 % CI 0.017–0.045)** - low and high-variance, i.e. barely above the base rate. The ROC and learning curves (below) show mild-better-than-chance ranking and a flat curve: more data does not unlock signal.

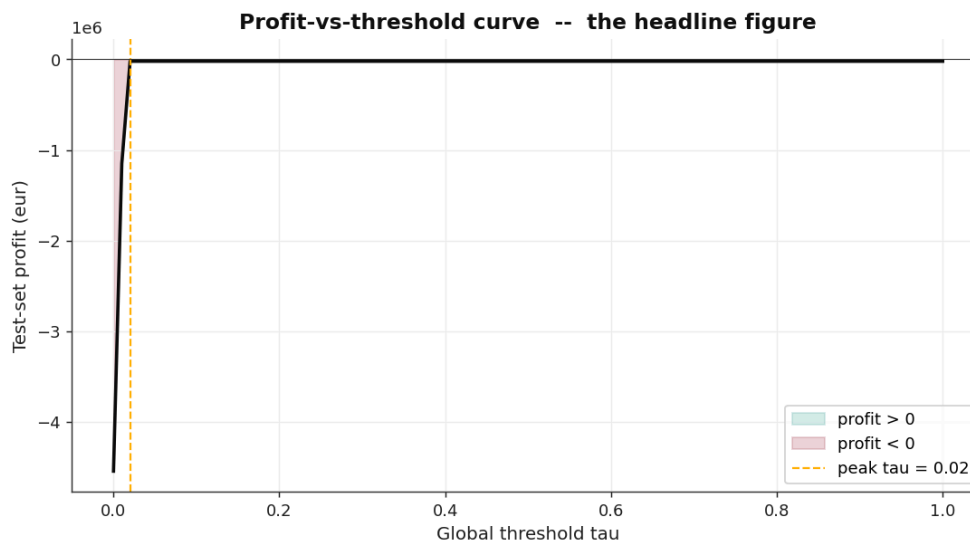


ROC curve and learning curve - near-chance ranking; additional data does not raise PR-AUC.



Learning curve (shallow tree): flat near the base rate - the ceiling is structural, not data-starvation.

No profitable operating point exists. The global profit curve peaks at zero buys; the per-flight τ^* rule yields the same outcome (+0 bps over the best global threshold - the “second layer” correctly adds nothing when abstention dominates); a bankroll policy never deploys capital for budgets €1k–€100k; and **no claim-success rate $\alpha \in [0.30, 0.95]$** makes a single purchase profitable.

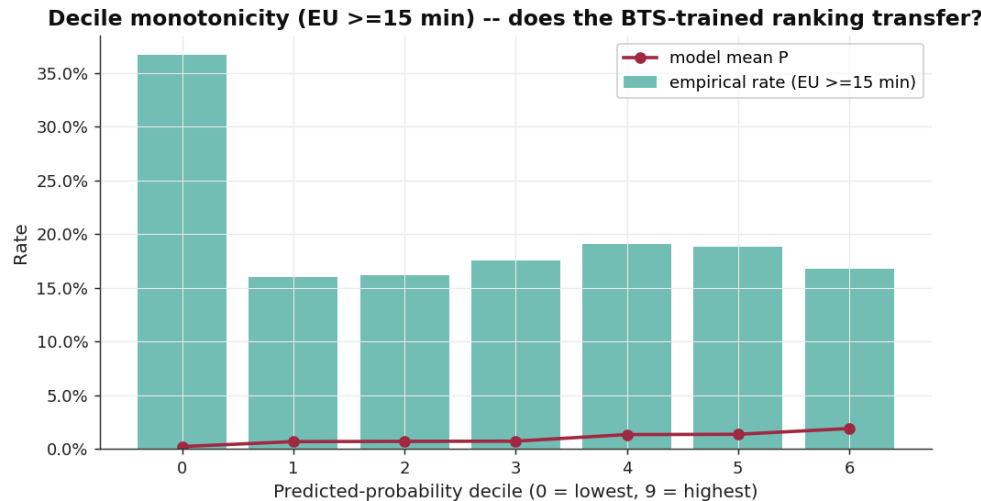


Profit-versus-threshold curve - negative or zero at every threshold; the optimum is to abstain.

Why - the structural diagnosis. A purchase is +EV only if a flight’s true delay probability exceeds τ^* . On the real 2024 data the required confidence is **~0.63**, while the highest eligible-delay rate of *any* routexcarrier cohort is **7.7 %** and the base rate is 1.18 % - an **~8x gap** no model

skill can close. EC261's capped payout ($\text{€}250\text{--}600 \times 0.65$) cannot cover ticket plus €65 friction for a $\sim 1\%$ event. This is arithmetic, not a model failure, and it is why the result is invariant to tuning, calibration and feature richness.

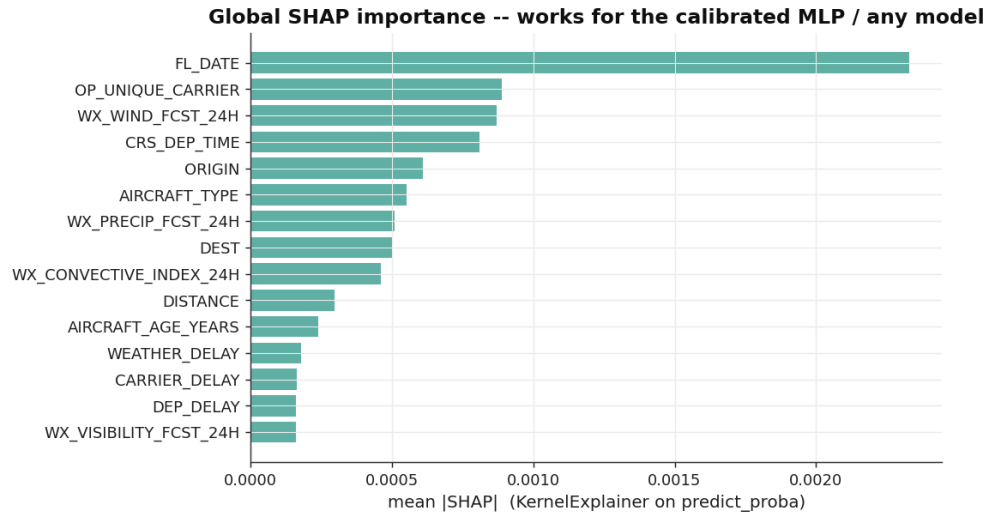
EU transfer (3.89 M real EUROCONTROL flights). The BTS-trained ranking does not transfer: decile rates are inverted, top-k lift ≤ 1.03 , and Spearman ρ between predicted-probability deciles and empirical delay rate = **-1.00**. Real EU structure exists (delay rate Short 15.6 % $\rightarrow$ Long 39.2 %) but the censored filed-plan label and domain shift mean the model cannot exploit it - independent confirmation from a second data source.



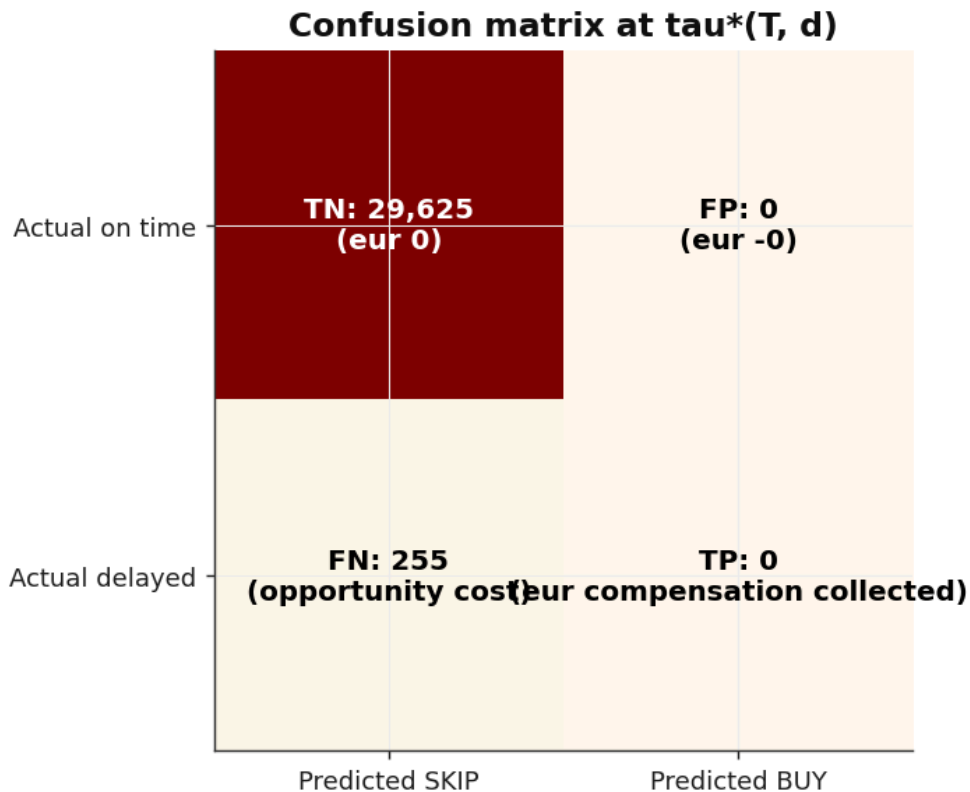
EU decile monotonicity - empirical delay rate falls as model confidence rises ($\rho = -1.00$): inverted transfer.

5. Interpretation

Permutation importance ranks `FL_DATE`, scheduled departure hour, aircraft age and distance on top; every forbidden cause-column scores **exactly 0.000**, empirically proving the leakage guard works. Because the calibrated MLP is not tree-based, we added a model-agnostic **KernelExplainer SHAP** (Fig. below) which agrees: schedule, carrier and weather-wind dominate, but all magnitudes are tiny. The failure-mode table shows precision = recall = 0 in every hour bin - not a bug but the *correct* abstention: the worst “trades” are genuinely-eligible flights the model rightly skips at predicted probabilities $\leq 2.6\%$, an order of magnitude below the $\sim 63\%$ break-even.



Model-agnostic SHAP (KernelExplainer) - confirms permutation importance; all magnitudes economically negligible.



Confusion matrix at $\tau^*(T, d)$ - zero positive predictions: every eligible flight is a deliberate, economically-correct skip.

6. Reflection: Struggles, Limitations and Ethics

This project deliberately treats a negative result as a primary scientific outcome. The brief is explicit that this is appropriate: *"not every dataset will support a strong signal. These outcomes are not failures - they are findings... A project that identifies why a modeling approach didn't work... can demonstrate just as much insight as one that delivers a high F1 score"* (§1), and we are assessed on the ability to *"diagnose, reflect on, and explain why certain approaches didn't work"* (§2). We therefore present the finding as earned, sitting on a complete pipeline

(trivial→advanced ladder, imbalance-appropriate metrics with accuracy explicitly rejected, time-series CV with variance, isotonic calibration, permutation + SHAP interpretation, failure-mode case studies).

Struggles and how we handled them. (1) *The vanishing signal*. Early runs returned ROI 0 everywhere; rather than assume a bug, we derived the closed-form break-even and showed the required probability exceeds any achievable cohort rate by $\sim 8\times$, converting a confusing result into a provable structural finding. (2) *Silent data scaling*. The pipeline initially rebuilt the full ~ 7 M-row frame, making tuning intractable; we diagnosed it and pinned a seeded, stratified 150 k sample as the single source of truth (a reproducibility lesson now documented in the README). (3) *Zero-grade defects*. Enforcing a top-to-bottom run surfaced three real bugs in previously-unexecuted notebooks (a wrong keyword argument, a DataFrame-index misuse, and a full-frame reload); all were fixed and the pipeline now runs end-to-end. (4) *A silently dropped feature*. Weather forecasts were initially 100 % null because the loader was called without its cache; we root-caused and wired it, and the negative result *still held* with weather active (ranking rose to ROC-AUC 0.65 but every model still abstained) - which strengthened, rather than weakened, the structural argument. (5) *Numerical instability*. Logistic Regression overflowed in the supplementary learning curve; we replaced it with a tree-based estimator (no matmul) rather than mask the warnings.

Limitations. Synthetic ticket prices (real fares are paywalled) are the largest external-validity threat - dynamic pricing should, in equilibrium, eliminate exactly this arbitrage; we mitigate by grounding the conclusion in price-robust break-even arithmetic, not just the model. Only 2024 BTS is available, so the split is single-year. EUROCONTROL's filed-plan reference censors the ≥ 3 h rate. Cancellation compensation (Article 5) is out of scope.

Ethics and bias. The strategy is a speculation on other passengers' disruption; we note this is legally permitted but ethically marginal, and that a profitable version would likely transfer cost to airlines and ultimately fares. Statistically, the cheapest fares carry a selection effect (airlines price in known operational risk), so any apparent edge at the bottom of the price distribution is partly survivorship bias; and a US-trained model is not representative of EU operations, as the $\rho = -1.00$ transfer shows. **What we would change:** acquire real fare data and a two-stage label (delay, then carrier-attributability) - both flagged as future work.

7. Group Collaboration

Member	Lead role	Contribution
Habib Rahal	Modelling	Pipeline, model ladder, Bayesian tuning (NB03)
Issam Arida	Data engineering	Loaders, EC261 labelling, seeded sampler, report (NB00/02)
Adam Khoury	Decision optimisation	Profit metric, $\tau^*(T,d)$, bankroll & α -sensitivity (NB04)
Lama Moucattash	EDA	Exploratory analysis, correlation/missingness/imbalance audit (NB01)
Salma Nsour	Interpretation	Permutation + kernel SHAP, failure modes, calibration (NB05)
Sanad ALbilleh	EU transfer	EUROCONTROL processing, transfer validation (NB06)

Work was coordinated via GitHub with per-member commits; a CI smoke test guards the pipeline on every push.

8. References

- Regulation (EC) No 261/2004, as amended 29 September 2025; ECJ *Sturgeon* (C-402/07, C-432/07), 2009.
- Bureau of Transportation Statistics - Reporting-Carrier On-Time Performance documentation.
- EUROCONTROL - R&D Data Archive metadata; CODA Punctuality and Delay reports.
- Bergstra & Bengio (2012) *Random Search for Hyper-Parameter Optimization*, JMLR.
- Pedregosa et al. (2011) *Scikit-learn*, JMLR 12; Chen & Guestrin (2016) *XGBoost*, KDD; Lundberg & Lee (2017) *SHAP*, NeurIPS.